

Project Initialization and Planning Phase

Date	2 July 2026
Team ID	xxxxxx
Project Title	OptiCrop – Smart Agricultural Production Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed OptiCrop – Smart Agricultural Production Engine is an AI and Machine Learning-based web application designed to help farmers select the most suitable crop based on soil nutrients and environmental conditions. The system analyzes parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall to provide accurate crop recommendations. This solution aims to improve agricultural productivity, reduce crop failure, and support data-driven farming decisions.

Project Overview	
Objective	To develop an intelligent crop recommendation system using Machine Learning that helps farmers choose the best crop based on soil nutrients and climatic conditions
Scope	The project provides crop recommendations through a user-friendly web application using Machine Learning. It can be extended with weather forecasting, fertilizer recommendation, and disease detection in future versions.
Problem Statement	

Description	Farmers often face difficulties in selecting the right crop due to changing weather conditions and lack of scientific guidance, leading to poor crop yield and financial losses.
Impact	Providing accurate crop recommendations helps improve productivity, reduce crop failure, increase farmers' income, and support sustainable agriculture.
Proposed Solution	
Approach	Use a trained Machine Learning model to analyze soil nutrient values (N, P, K), temperature, humidity, pH, and rainfall, then recommend the most suitable crop through a Flask-based web application.
Key Features	• AI-based crop recommendation • User-friendly Flask web interface • Fast prediction using a trained ML model • Accurate recommendations based on soil and weather conditions • Easy input and instant prediction results

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU	Intel Core i5 (or equivalent) CPU
Memory	RAM	8 GB RAM
Storage	Disk	256 GB SSD or above

Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	Scikit-learn, Pandas, NumPy, Joblib, Matplotlib
Development Environment	IDE	Visual Studio Code, Jupyter Notebook
Data		
Data	Source, size, format	Kaggle